

The Escape Is Selectable: A Text-Gated Router for Composing Sensor Bricks

Franci Penov
kortexa.ai
francip@kortexa.ai

2026-08-04

Abstract

This note expands on paper 7 (*Zero Weight-Space Interference Is Not Enough* [1]), which closed additive composition of sensor bricks on a frozen language model and left a constructive existence proof with an open problem: oracle selection recovers every brick exactly (8/8 seeds), but the only learned mechanism for engaging it — a task-conditioned router — must be *given* the task identity and escapes its collapse basin in only 6 of 8 seeds. We replace the learned router with a frozen text classifier: at evaluation, a lane selector reads the eval item’s own text surface and emits a hard one-hot over bricks, composed exactly as the oracle’s. Nothing trains through the LM loss, so there is no collapse basin. Under a preregistered two-round protocol, both selector arms — a 175,107-trainable-parameter frozen 350M encoder with a tuned head, and a 39,235-parameter byte-CNN — clear the registered roll-up at **8/8 seeds with retention 1.000 in both modalities**, against the cited 6/8 for the learned router. Both of paper 7’s load-bearing caveats fall at this scope. Three discounts travel with the claim: the dispatch requires in-domain supervision, pretraining buys nothing at the wired level, and the leakage exclusion is item-level because a text-level rule would empty both task lanes.

1 Why this note exists

Paper 7’s constructive section ended one step short of a method. Selection composes exactly: under **gate-oracle**, each brick’s out-of-objective probe returns to its single-brick score at every one of eight seeds. But the learned task-conditioned router that engages selection carries two caveats at paragraph prominence: it is handed the task identity through a learned embedding (the sensor-feature stream carries no routing signal by construction — the feature-only null’s between-modality L_1 gap is exactly 0.000 at all eight seeds), and its escape from the collapse basin is seed-determined (6 of 8, always starving audio). The registered follow-up was to make the escape *selectable*. A substrate bake-off (recorded in the program log) found that prompt-lane classification is saturable by small frozen encoders where generation steering fails. This note wires that selector in and runs the experiment the open problem asked for.

2 The experiment

One new gate mode. **gate-selector** evaluates the same composition as **gate-oracle** with one substitution: the one-hot over bricks comes from a frozen lane classifier reading the eval item’s own text surface, not from ground truth. No task embedding exists, no gate steps run, and nothing

trains through the LM loss — so seed variation can come only from the bricks and the eval draw, which the oracle column already shows do not decide anything. Two arms share the wiring: the registered substrate (LFM2.5-Encoder-350M, frozen, LoRA rank 4 on attention plus a linear head; 175,107 trainable parameters) and a from-scratch byte-CNN (39,235 parameters) as the pretraining control. Each selector is trained once (seed 314159) and frozen across all eight harness seeds (42, 1042, 2042, 3042, 4042, 5042, 6042, 7042), at the paper-7 Arm-5 configuration verbatim: three bricks (IMU, audio, vision) on disjoint 768-dimensional coefficient blocks over a shared frozen basis on LFM 2.5 230M, 64 eval items per modality per seed.

The integrity check. On any eval item where the selector’s argmax equals the true brick, the composed weights must equal `gate-oracle`’s for that item bit-for-bit, and the probe rank must match; a violation raises rather than reports. Across both rounds, 2,344 correctly-routed items were checked and 2,344 matched — a rate of 100.0%, by construction. Every gap between a selector column and the oracle column is therefore selector error and nothing else: retention is pinned to the ceiling wherever the classifier is right, and the experiment’s only free quantity is classifier accuracy.

Leakage rule. The eight seeds’ eval items are enumerable up front: 1,024 items over 56 distinct text surfaces. Round 1 excluded every training item whose text matched any eval surface — 12,299 of 25,081 pool items (audio 2,000, imu 10,299) — and the selectors never saw an eval item.

Two rounds, registered in advance. The preregistration named three readings and permitted exactly one iteration, gated mechanically: a launcher script renders the round-1 verdict from the artifacts and refuses to run round 2 unless it reads *selector-limited* — retention failing exactly where the selector was wrong.

3 Result

gate	round 1	round 2
<code>gate-oracle</code> (given identity; ceiling)	8/8	8/8
<code>gate-selector</code> (encoder, text-gated)	0/8	8/8
<code>gate-selector-cnn</code> (byte-CNN, text-gated)	0/8	8/8
<code>gate-task-lr-*</code> (learned router, cited from paper 7)	6/8	

Table 1: Seeds (of 8) whose retention clears 80% in *every* modality. Round-2 selector retention is 1.000 in both modalities at all eight seeds, in both arms. The learned-router row is cited from paper 7’s artifacts (escaped at seeds 42, 2042, 4042, 5042, 6042, 7042) and was not re-run.

Round 1 obtained the selector-limited reading. The bake-off’s selectors, trained on a foreign rendering of the same lanes, do not transfer to the harness’s text surface: the encoder scores 57.8% on audio and 0.0% on IMU eval items, the byte-CNN 0.0% on both, and both arms roll up 0/8. Retention tracks accuracy mechanically (Table 2): encoder audio retention spans 52.4%–77.3% at 57.8% accuracy, and IMU retention sits at 17.9%–25.0% — which is the `others_only` floor, because a wrong one-hot composes exactly the `others_only` control. Selector accuracy has zero seed variance by construction: the eval text multiset is seed-invariant, so a text-only gate is *seed-constant* — the opposite failure geometry from the learned router’s bistability.

Round 2 saturates. Retrained on the harness’s own train-split streams (item-level exclusion; for IMU no eval *subject* is ever seen), both arms score 64/64 on the eval texts of both modalities at every seed, and retention is 1.000 in both modalities at all eight seeds — matching the oracle column cell for cell. The single-brick references the retention is measured against are unchanged

arm	modality	selector accuracy	retention across seeds
encoder	audio	57.8%	52.4%–77.3%
encoder	imu	0.0%	17.9%–25.0%
byte-CNN	audio	0.0%	4.2%–6.7%
byte-CNN	imu	0.0%	17.9%–25.0%

Table 2: Round 1: the coupling the integrity check enforces. Retention is a mechanical function of selector accuracy; nothing else in the harness moves.

throughout (seed 42: IMU 0.750, audio 0.297, reproducing paper 7’s Arm-5 comparator to the digit).

Both of paper 7’s caveats fall at this scope. The gate is not given task identity — it reads the question — and there is no bistability, because there is no basin: nothing is trained through the LM loss, and the escape-rate column for a text gate is a constant of the text, not a draw of the seed.

4 What does not follow

Dispatch needs in-domain supervision. Round 1 is a real negative: classifiers at 0.997–1.000 held-out accuracy on their own training surface transferred at 57.8%/0.0% (encoder) and 0/0 (byte-CNN) to the harness’s surface. The direction is symmetric — the round-2 encoder drops to 95.3% on the template-stripped surface it no longer trains on, the round-2 byte-CNN to 67.2%. Nothing here demonstrates zero-shot lane inference; the claim is that *supervised* dispatch, trained on the deployment surface, engages selection exactly.

Pretraining buys nothing at the wired level. The 39,235-parameter byte-CNN matches the 350M encoder in every cell of Table 1. At this scope the dispatch is a surface-features problem, and the encoder’s advantages live off-surface (transfer, calibration), not in the wired result. A method claim built on the encoder specifically would overstate what this experiment shows.

The exclusion is item-level, not text-level. The registered text-level rule cannot apply to round 2: excluding every training item whose text matches an eval surface would remove 7,352 IMU and 1,600 audio items — both task lanes in their entirety, recorded in the artifact as `text_rule_would_empty_lanes`. Round 2’s selectors have therefore seen the eval *label words* in other items — never an eval item, and for IMU never an eval subject. This is load-bearing for how strong “never seen” is, and it is why the round-1/round-2 distinction is reported rather than collapsed.

Scope. One model (LFM2.5230M), one brick geometry (disjoint-768 over a shared frozen basis), two powered modalities, $n = 64$ eval items per cell (± 0.091 binomial half-width at 95%); each selector was trained once at seed 314159 (MPS training is not bit-reproducible, so the selector weights, unlike the splits, are not reproducible artifacts); and the roll-up threshold is inherited from paper 7, not chosen here.

5 Consequence for the program

Paper 7’s open problem is closed at this scope, and its constructive section upgrades from existence proof to method contingent on scale: selection composes, and a text-gated selector engages it without task identity and without bistability. The composition rail ladders before it lifts — the

next rungs are selection over real hypernetwork bricks at 0.8B and above, and more task bricks, where the collapse mechanism predicts learned routing degrades and a supervised selector should not care.

References

- [1] F. Penov. Zero Weight-Space Interference Is Not Enough: Behavioral Probes for Composing Sensor Bricks on a Frozen Language Model. Preprint, 2026. <https://github.com/kortexa-ai/legolm.basis>.
- [2] F. Penov. Two Regimes of Compressed Conditioning. Preprint, 2026. <https://github.com/kortexa-ai/legolm.basis>.
- [3] Liquid AI. LFM2.5 model family. Model releases, 2025–2026. <https://huggingface.co/LiquidAI>.